

Assignment Day 6 - Strings

1. Count Vowels in a String

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str = "Hello World";
    int count = 0;

    for(char ch : str) {
        if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
           ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
            count++;
    }

    cout << "Vowels = " << count;
    return 0;
}
```

Input: Hello World

Output: Vowels = 3

2. Reverse String

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str = "Hello";

    for(int i = str.length()-1; i >= 0; i--)
        cout << str[i];

    return 0;
}
```

Input: Hello

Output: olleH

3. Palindrome Check

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str = "madam";
    string rev = "";

    for(int i = str.length()-1; i >= 0; i--)
        rev += str[i];

    if(str == rev)
        cout << "Palindrome";
    else
        cout << "Not Palindrome";

    return 0;
}
```

Input: madam

Output: Palindrome

4. Count Words in a String

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str = "I Love C Plus Plus";
    int words = 1;

    for(char ch : str) {
        if(ch == ' ')
            words++;
    }

    cout << "Words = " << words;
    return 0;
}
```

Input: I Love C Plus Plus

Output: Words = 5

5. Lowercase to Uppercase

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    string str = "hello world";

    for(char &ch : str)
        ch = toupper(ch);

    cout << str;
    return 0;
}
```

Input: hello world

Output: HELLO WORLD

6. Anagram Check

```
#include <iostream>
#include <algorithm>
#include <string>
using namespace std;

int main() {
    string str1 = "listen";
    string str2 = "silent";

    sort(str1.begin(), str1.end());
    sort(str2.begin(), str2.end());

    if(str1 == str2)
        cout << "Anagram";
    else
        cout << "Not Anagram";

    return 0;
}
```

Input: listen, silent

Output: Anagram